

# MicroserviceBase

**v. 2.0.0**

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# Chapter 1

## Introduction

### 1.1 Introduction

**MicroserviceBase** is a Python framework for building, managing, and orchestrating microservices with RabbitMQ-based communication.

It provides:

- **Microservice Framework:** Base classes for creating services with RPC and pub/sub patterns
- **Service Registry:** Dynamic service registration and discovery via RabbitMQ exchange
- **FastAPI Bridge:** REST API and WebSocket gateway connecting GUI to microservices
- **Manager GUI:** Electron and browser-based dashboard for monitoring and controlling services
- **Local Process Hub:** Process lifecycle management with graceful shutdown
- **Hexagonal Architecture:** Clean separation via ports and adapters pattern

#### 1.1.1 Target Audience

This package is designed for developers who need to:

- Build microservices that communicate via RabbitMQ message broker
- Monitor and manage services through a graphical dashboard
- Manage local service processes with start, stop, and import capabilities
- Deploy a standalone desktop application for service management

#### 1.1.2 Prerequisites

- Python 3.10 or higher
- RabbitMQ server (message broker)
- pika package (RabbitMQ client for Python)
- FastAPI and uvicorn (for the bridge)
- Node.js and npm (for GUI development)

The sources of **MicroserviceBase** are available in [GitHub](#).

Information about how to install the **MicroserviceBase** can be found in the [README](#).

# Chapter 2

## Description

### 2.1 Description

#### 2.1.1 Overview

MicroserviceBase is a Python framework for building microservices that communicate through RabbitMQ, with a built-in GUI for monitoring, a local process hub for lifecycle management, and a service registry for dynamic discovery. The system follows hexagonal architecture principles, ensuring testability, maintainability, and extensibility.

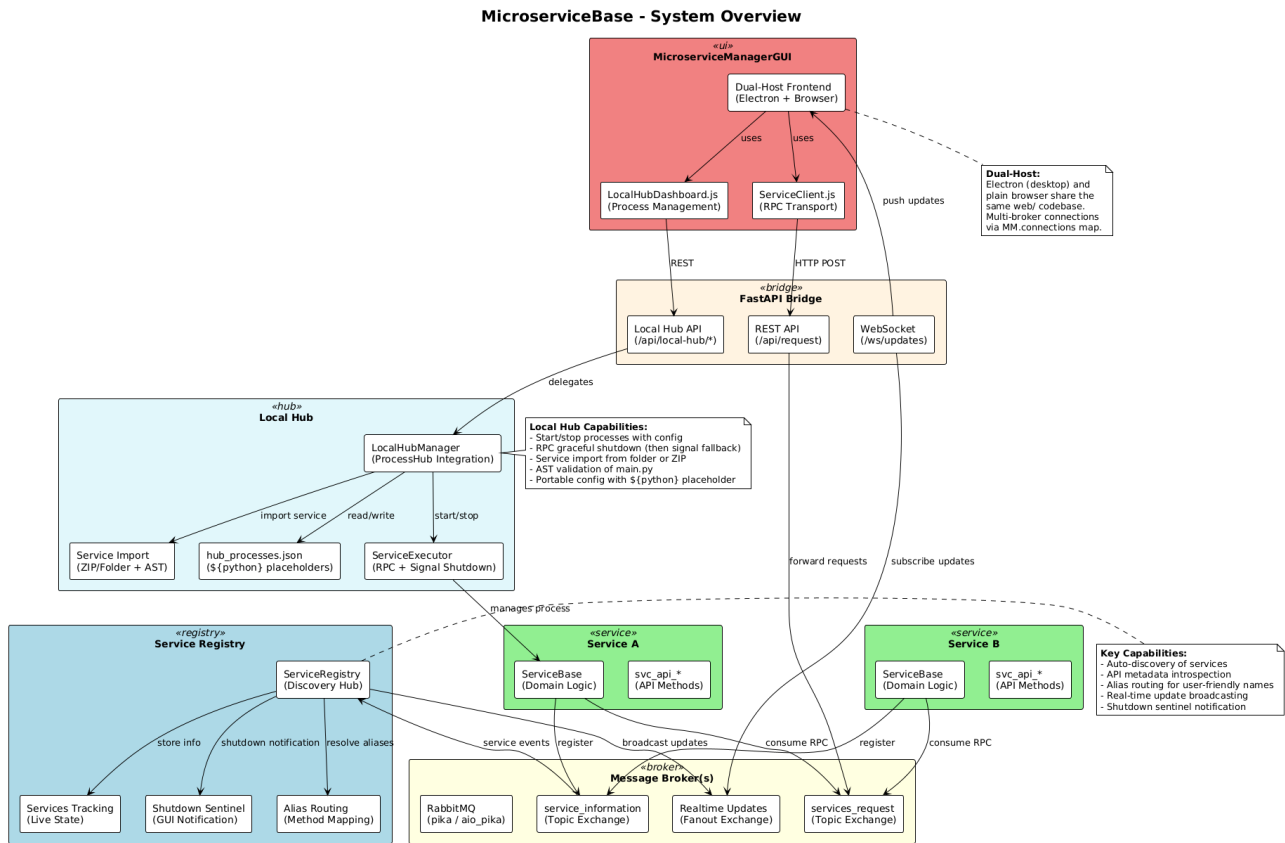


Figure 2.1: MicroserviceBase System Overview

#### Key Features

- **Microservice Framework:** Base classes for creating services with RPC and pub/sub patterns
- **Service Registry:** Dynamic service registration and discovery via RabbitMQ exchange
- **FastAPI Bridge:** REST API and WebSocket gateway connecting GUI to microservices
- **Manager GUI:** Electron + browser-based dashboard for monitoring and controlling services

- **Local Process Hub:** Start, stop, and manage service processes with graceful shutdown
- **Multi-Broker Support:** Connect to multiple RabbitMQ brokers simultaneously
- **Service Import:** Import microservice packages (folder or zip) into the local hub
- **Service Creator:** Interactive wizard for scaffolding new microservices
- **Standalone Installer:** Package as Windows desktop application (DevAtServGUI)

## Design Philosophy

MicroserviceBase follows several key design principles:

1. **Hexagonal Architecture:** Domain logic is isolated from external concerns via ports and adapters
2. **Factory Pattern:** All components are created through factories, enabling easy swapping
3. **Dual Hosting:** GUI works in both Electron (desktop) and browser modes
4. **Graceful Shutdown:** Proper process cleanup on Windows using CTRL\_BREAK\_EVENT signals
5. **Config Persistence:** Placeholder-based configuration that works across environments

### 2.1.2 Architecture

MicroserviceBase is organized into distinct layers following hexagonal architecture:

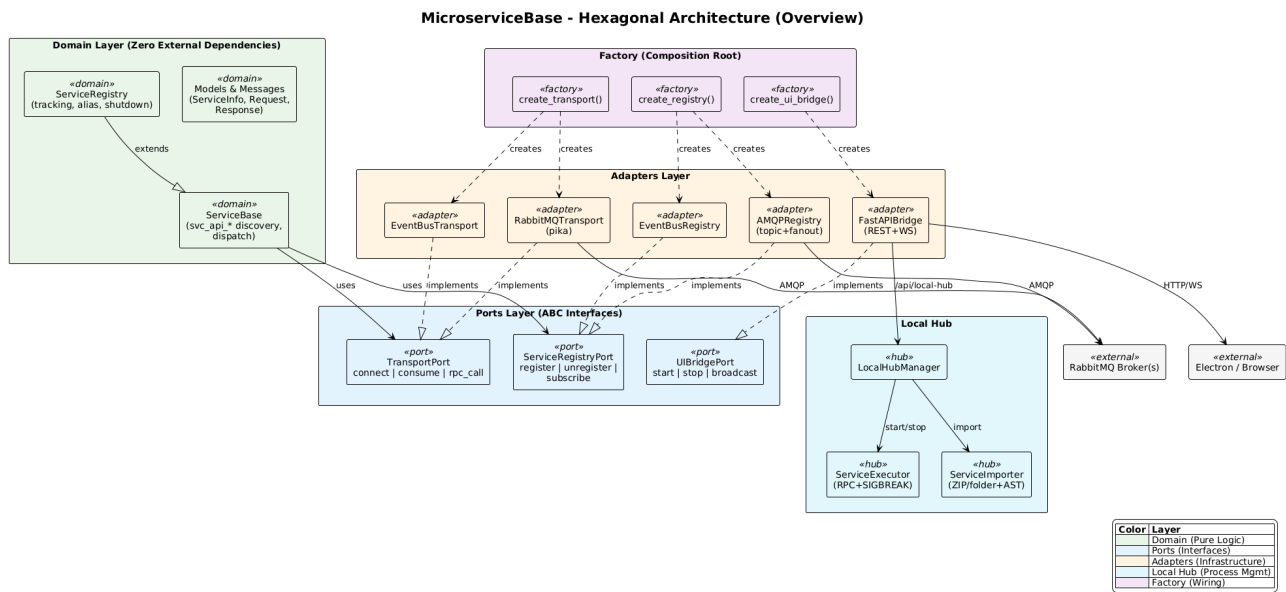


Figure 2.2: Hexagonal Architecture Overview

The detailed architecture with method signatures and implementation specifics:

#### Layer Responsibilities

**Domain Layer** Core business logic: `ServiceBase` (base class for microservices), `ServiceRegistry` (service registration and discovery), and `Factory` (component creation).

**Ports Layer** Abstract interfaces defining contracts: `TransportPort` (message passing), `RegistryPort` (service registration), `UIBridgePort` (GUI communication).

**Adapters Layer** Concrete implementations:

- **RabbitMQ Transport** – RPC calls and message consumption via pika
- **RabbitMQ Registry** – Service registration via exchange broadcasting

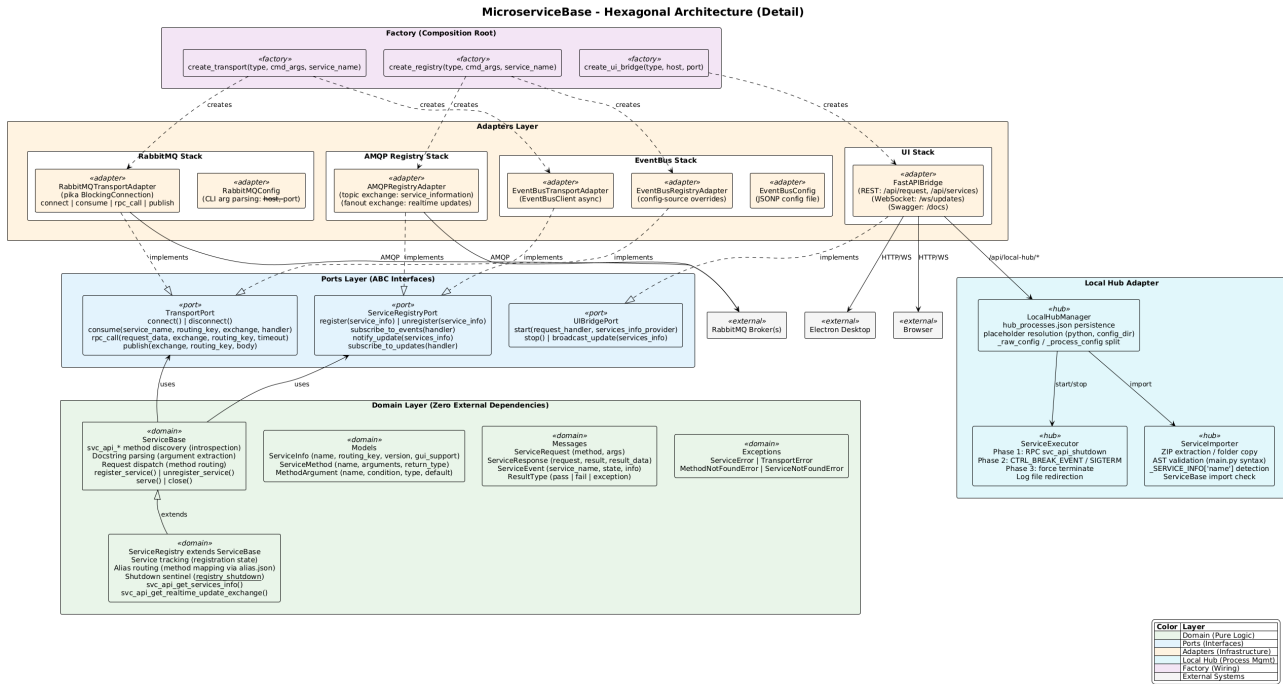


Figure 2.3: Hexagonal Architecture Detail

- FastAPI Bridge – REST API, WebSocket, and Local Hub endpoints
- Local Hub Manager – Process lifecycle management
- Service Executor – Process execution with platform-specific shutdown

GUI Layer MicroserviceManagerGUI with dual hosting:

- Electron – Desktop wrapper with contextBridge preload
- Web – Pure browser-compatible HTML/CSS/JS
- Service Plugins – Dynamically loaded per-service UI components

### 2.1.3 Components

#### ServiceBase

The base class for all microservices provides:

- API method discovery via `svc_api_` prefix convention
- Automatic service registration with the registry
- Graceful shutdown with `unregister_service()` on exit
- Service info exposure for discovery via `_SERVICE_INFO` dict
- Docstring-based API documentation generation

```
from MicroserviceBase.domain.service_base import ServiceBase
from MicroserviceBase.factory import create_transport, create_registry
```

```
class MyService(ServiceBase):
    _SERVICE_INFO = {
        'name': 'MyService',
        'routing_key': 'service.myservice',
        'version': '1.0.0',
        'gui_support': False,
        'methods': [], 'methods_info': {},
    }
```

```

def svc_api_my_method(self, arg1, arg2):
    """My API method."""
    return some_result

transport = create_transport('rabbitmq',
    cmd_args=['--host', 'localhost', '--port', '5672'],
    service_name='MyService')
registry = create_registry('rabbitmq',
    cmd_args=['--host', 'localhost', '--port', '5672'],
    service_name='MyService')

service = MyService(transport=transport, registry=registry)
service.register_service()
service.serve()

```

## Service Registry

The registry manages dynamic service discovery:

- Services register on startup with their method list and metadata
- Registry broadcasts updates to all subscribers via RabbitMQ exchange
- Shutdown notification via `__registry_shutdown__` sentinel
- GUI receives real-time updates via WebSocket

## FastAPI Bridge

The bridge connects the GUI to microservices:

**POST /api/request** RPC call forwarding to services

**GET /api/services** Registered service list

**WS /ws/updates** Real-time service update stream

**/api/local-hub/\*** Local process hub management endpoints

**GET /docs** Auto-generated Swagger API documentation

## Local Hub Manager

The Local Hub provides process lifecycle management:

- Start, stop, and monitor local service processes
- Process configuration via `hub_processes.json` with placeholder support
- Service import from folders or zip archives
- Service executor with graceful shutdown using `CTRL_BREAK_EVENT` on Windows
- Log file management per process

## 2.1.4 Communication Patterns

### RPC via RabbitMQ

Services communicate using RPC (Remote Procedure Call) pattern through RabbitMQ. The following diagram shows the complete communication flow between a service, the Service Registry, and a client, including all exchanges and routing keys used:

```

from MicroserviceBase.domain.service_base import ServiceBase

# Client sends request
request_data = ServiceBase.create_request_data('svc_api_add', [1, 2])
response = transport.rpc_call(
    request_data,
    exchange_name='services_request',
    routing_key='service.calculator'
)
# response = {"request": "svc_api_add", "result": "pass",
#            "result_data": 3}

```

### Service Registration

Services register themselves via the registry exchange. The `ServiceBase` class handles this automatically using the `_SERVICE_INFO` dict:

```

# ServiceBase auto-discovers svc_api_* methods and builds
# the service info dict, then publishes it:
service.register_service()

# On shutdown, unregister to notify the registry:
service.unregister_service()

```

### Multi-Broker Architecture

The GUI supports connecting to multiple RabbitMQ brokers simultaneously:

- `MM.connections` map tracks per-broker state
- `serviceToBroker` reverse lookup for request routing
- Session persistence via `sessionStorage`
- Progressive disclosure: broker sections shown only when multiple brokers connected

## 2.1.5 GUI Architecture

### Dual Hosting

The `MicroserviceManagerGUI` runs in two modes:

**Electron Mode** Desktop application with native process management. Uses `contextBridge` preload for secure Node.js access. Can spawn and manage the FastAPI bridge process directly.

**Browser Mode** Access via `http://localhost:1112` when the bridge is running. Pure HTML/CSS/JS with no Node.js dependencies. Uses the same `window.MicroserviceManager` namespace.

### Service Plugins

Each microservice can provide a custom GUI plugin:

- HTML file for UI layout
- JS file for logic (uses `const MM = window.MicroserviceManager;`)
- Plugins loaded dynamically into `web/services/`
- In packaged mode, plugins extracted to writable `%APPDATA%` directory

## Standalone Installer

The GUI can be packaged as a standalone Windows application using Electron Builder:

- NSIS installer for Windows (`build.bat`), Linux packages (`build.sh`)
- Read-only scripts in `resources/python/`
- Writable user data in `%APPDATA%/devatservgui/`
- First-run initialization copies templates and scripts
- Bridge process managed via PID file with detached mode

### 2.1.6 Process Configuration

Processes are configured via `hub-processes.json`:

```
{
  "ServiceRegistry": {
    "script": "${config_dir}/start_registry.py",
    "args": ["--config", "${config_dir}/config.json"],
    "process_name": "ServiceRegistry",
    "wait_time": 2.0
  },
  "MyService": {
    "script": "${python}",
    "args": ["-m", "MyService"],
    "cwd": "${config_dir}/services",
    "process_name": "MyService",
    "wait_time": 1.0
  }
}
```

## Placeholders

`${python}` Resolves to `sys.executable` at runtime

`${config_dir}` Resolves to the directory containing `hub-processes.json`

Placeholders are preserved in `_raw_config` during save operations, allowing the same configuration file to work across different environments.

### 2.1.7 Windows Process Lifecycle

On Windows, process shutdown requires special handling:

- `proc.terminate()` calls `TerminateProcess` which allows no cleanup
- Service Executor uses `CTRL_BREAK_EVENT` for graceful shutdown
- `signal.SIGBREAK` handler converts to `KeyboardInterrupt` for Python cleanup
- Pika's `start_consuming()` replaced with `process_data_events(time_limit=1)` loop for interruptible consumption
- `subprocess.PIPE` blocking prevented by using `FileHandler` instead of `StreamHandler` for logging in subprocess mode

### 2.1.8 Installation

#### From Source

```
git clone https://github.com/test-fullautomation/python-microservice-base.git
cd python-microservice-base
pip install .
```

```
# Development mode
pip install -e .
```

#### GUI Setup

```
cd MicroserviceBase/MicroserviceManagerGUI
npm install
npm start
```

### 2.1.9 Quick Start

#### Start Service Registry

```
python MicroserviceBase/MicroserviceManagerGUI/python/start_registry.py \
    --host localhost --port 5672
```

#### Start FastAPI Bridge

```
python MicroserviceBase/MicroserviceManagerGUI/python/start_bridge.py \
    --host localhost --port 5672 --bridge-port 1112
```

#### Create a Service

```
import sys
from MicroserviceBase.domain.service_base import ServiceBase
from MicroserviceBase.factory import create_transport, create_registry

class CalculatorService(ServiceBase):
    _SERVICE_INFO = {
        'name': 'Calculator',
        'description': 'A simple calculator service.',
        'version': '1.0.0',
        'routing_key': 'service.calculator',
        'gui_support': False,
        'methods': [], 'methods_info': {},
    }

    def svc_api_add(self, a, b):
        """Add two numbers."""
        return int(a) + int(b)

transport = create_transport('rabbitmq',
    cmd_args=sys.argv[1:], service_name='Calculator')
registry = create_registry('rabbitmq',
    cmd_args=sys.argv[1:], service_name='Calculator')

service = CalculatorService(transport=transport, registry=registry)
service.register_service()
service.serve()
```

### 2.1.10 Diagrams Reference

The following PlantUML diagrams are available in the `docs/diagrams/` folder:

**overview.puml** System overview showing services, broker, registry, bridge, and GUI

**architecture.puml** Hexagonal architecture with domain, ports, and adapters

**component.puml** Component diagram with package dependencies

**class\_domain.puml** Class diagram for domain layer

**class\_ports.puml** Class diagram for port interfaces

**class\_adapters.puml** Class diagram for adapter implementations

**gui\_architecture.puml** GUI dual-host architecture (Electron + browser)

**sequence\_rpc.puml** RPC call sequence from GUI to service

**sequence\_registration.puml** Service registration flow

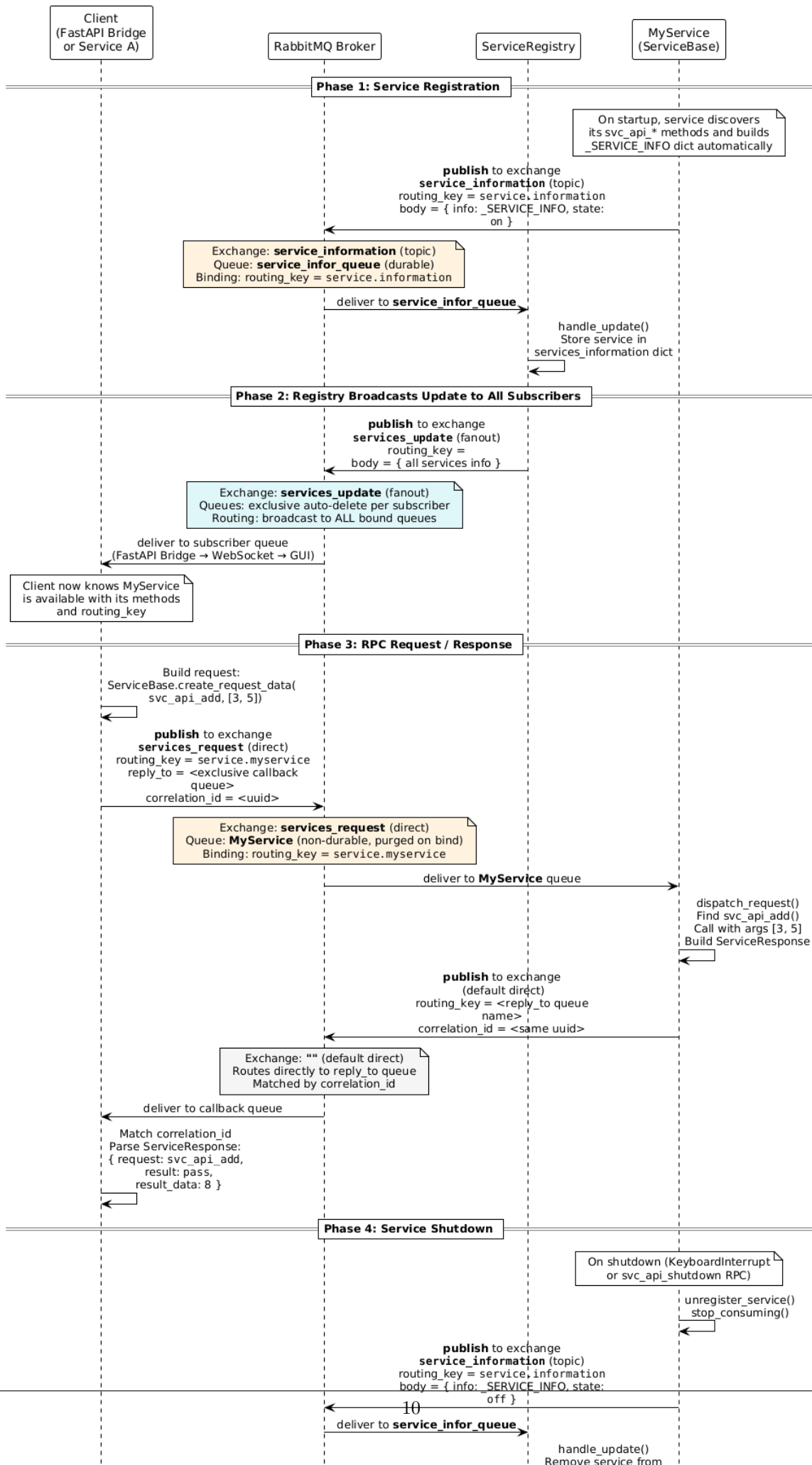
**sequence\_alias.puml** Service alias resolution sequence

**sequence\_shutdown.puml** Two-phase graceful shutdown sequence

**sequence\_service\_import.puml** Service import flow (GUI to hub)

**state\_process\_lifecycle.puml** Process state machine

## Service Communication Flow (Exchanges, Queues, and Routing Keys)



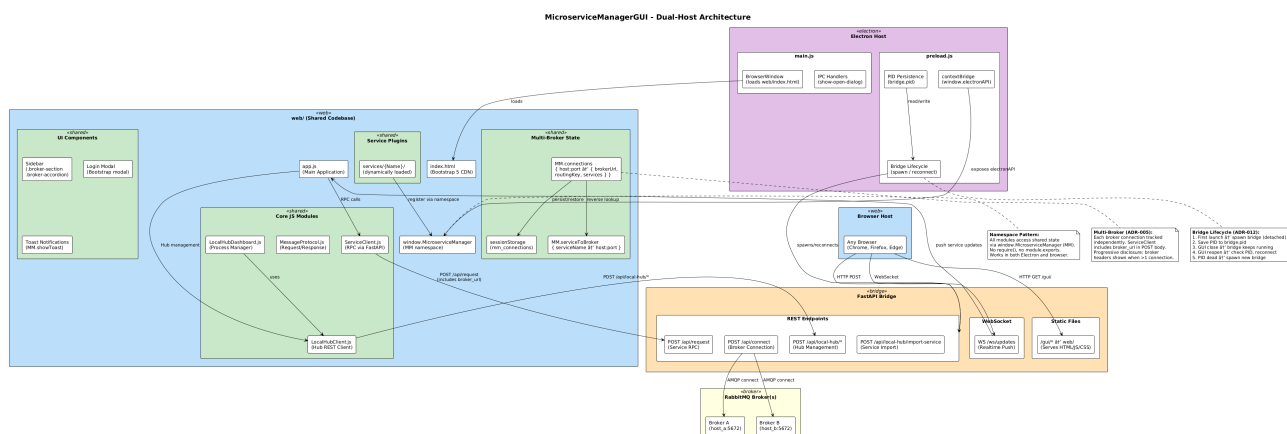


Figure 2.5: GUI Dual-Host Architecture (Electron + Browser)

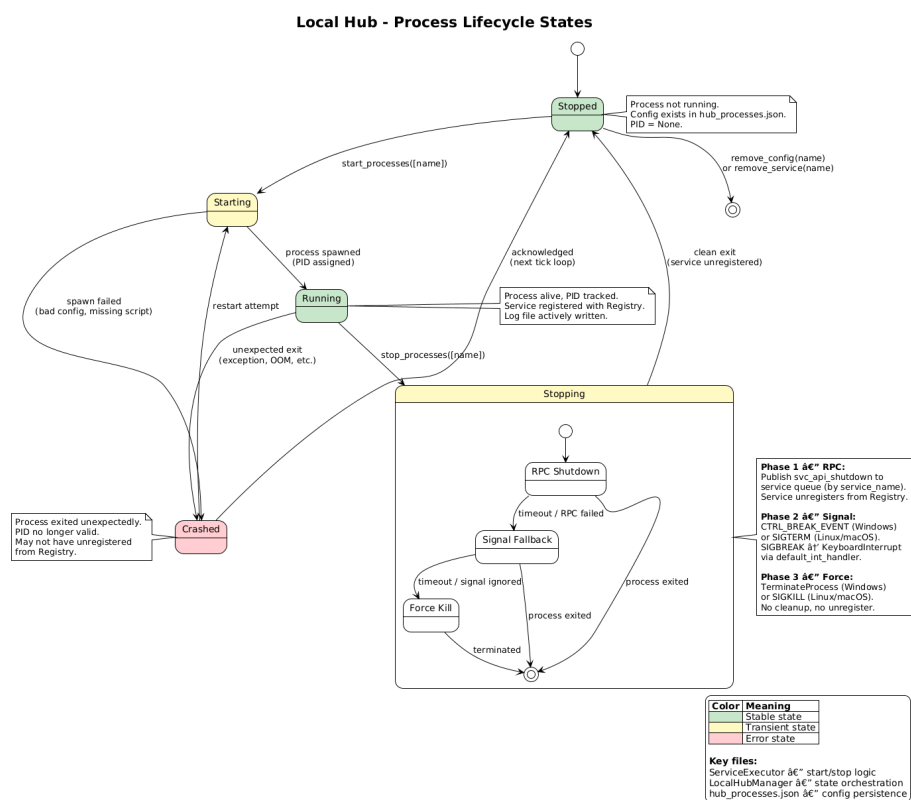


Figure 2.6: Local Hub Process Lifecycle States

## Chapter 3

# launcher.py

FastAPI Bridge launcher.

Starts the FastAPI Bridge (REST/WS gateway) that connects the MicroserviceManagerGUI to microservices via RabbitMQ.

### 3.1 Function: parse\_args

### 3.2 Function: main

## Chapter 4

# ClewareAccessHelper.py

### 4.1 Class: ClewareAccessHelper

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.ClewareAccessHelper
↳ import ClewareAccessHelper
```

ClewareAccessHelper class acts as a proxy class in Proxy pattern and forward the request to the real helper depend on platform. microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-import-modules-from-paths -----

Import all modules from given paths.

**Arguments:**

- paths  
/ Condition: required / Type: list /  
List of paths to import modules from.

#### 4.1.1 Method: load\_config

Load the user config port name for USB access. Returns: None microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-get-config -----

Get Cleware ports' config. Returns: Dictionary of ports' setting. microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-set-config -----

Save Cleware ports' config to file. Args: config\_json: data to be saved.

**Returns:** res: saved result.

- 4.1.2 Method: `init_cleware`
- 4.1.3 Method: `open_cleware`
- 4.1.4 Method: `close_cleware`
- 4.1.5 Method: `get_handle`
- 4.1.6 Method: `set_value`
- 4.1.7 Method: `get_value`
- 4.1.8 Method: `set_switch`
- 4.1.9 Method: `set_switch_by_port_name`
- 4.1.10 Method: `get_switch`
- 4.1.11 Method: `get_version`
- 4.1.12 Method: `get_usb_type`
- 4.1.13 Method: `get_serial_number`
- 4.1.14 Method: `valid_ser_number`
- 4.1.15 Method: `get_hw_version`
- 4.1.16 Method: `is_ampel`
- 4.1.17 Method: `iox`
- 4.1.18 Method: `get_all_devices_state`

## Chapter 5

# ClewareAccessHelperAbs.py

### 5.1 Class: ClewareAccessHelperAbs

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.ClewareAccessHelperAbs  
↳ import ClewareAccessHelperAbs
```

ClewareAccessHelperAbs acts as a abstract class for the real Helper in Proxy Pattern. microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-clewareservice-clewareaccesshelperabs-clewareaccesshelperabs  
init-cleware -----  
-----

5.1.1 Method: open\_cleware

5.1.2 Method: close\_cleware

5.1.3 Method: set\_value

5.1.4 Method: get\_value

5.1.5 Method: set\_switch

5.1.6 Method: get\_switch

5.1.7 Method: get\_handle

5.1.8 Method: get\_version

5.1.9 Method: get\_usb\_type

5.1.10 Method: get\_usb\_type

5.1.11 Method: get\_serial\_number

5.1.12 Method: valid\_ser\_number

5.1.13 Method: get\_hw\_version

5.1.14 Method: is\_ampel

5.1.15 Method: iox

## Chapter 6

# ClewareAccessHelperLinux.py

### 6.1 Class: ClewareAccessHelperLinux

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.ClewareAccessHelperLinux  
↳ import ClewareAccessHelperLinux
```

**ClewareAccessHelperLinux acts as the real object on Linux platform in Proxy Pattern**

- This is the wrapper class for the functions provided by C/C++ source for Linux.
- The C/C++ source for Linux is provided by Cleware manufacturer at : [http://www.cleware-shop.de/Downloads\\_EN](http://www.cleware-shop.de/Downloads_EN)
- The C/C++ source for Linux is modified by Cuong Nguyen (RBVH/ENG22) for support python to load dynamic library on Linux.

- 6.1.1 Method: `init_cleware`
- 6.1.2 Method: `open_cleware`
- 6.1.3 Method: `close_cleware`
- 6.1.4 Method: `get_handle`
- 6.1.5 Method: `set_value`
- 6.1.6 Method: `get_value`
- 6.1.7 Method: `set_switch`
- 6.1.8 Method: `get_switch`
- 6.1.9 Method: `get_version`
- 6.1.10 Method: `get_usb_type`
- 6.1.11 Method: `get_serial_number`
- 6.1.12 Method: `valid_ser_number`
- 6.1.13 Method: `get_hw_version`
- 6.1.14 Method: `is_ampel`
- 6.1.15 Method: `iox`
- 6.1.16 Method: `get_all_sw_state`

## Chapter 7

# ClewareAccessHelperWindows.py

### 7.1 Class: ClewareAccessHelperWindows

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.ClewareAccessHelperWindows  
↳ import ClewareAccessHelperWindows
```

**ClewareAccessHelperWindows acts as the real object on Windows platform in Proxy Pattern**

- This is the wrapper class for the functions provided by USBaccess.dll.
- The USBaccess.dll is provided by Cleware manufacturer at : [http://www.cleware-shop.de/Downloads\\_EN](http://www.cleware-shop.de/Downloads_EN)

**7.1.1 Method: init\_cleware**

**7.1.2 Method: open\_cleware**

**7.1.3 Method: close\_cleware**

**7.1.4 Method: get\_handle**

**7.1.5 Method: set\_value**

**7.1.6 Method: get\_value**

**7.1.7 Method: set\_switch**

**7.1.8 Method: get\_switch**

**7.1.9 Method: get\_version**

**7.1.10 Method: get\_usb\_type**

**7.1.11 Method: get\_serial\_number**

**7.1.12 Method: valid\_ser\_number**

**7.1.13 Method: get\_hw\_version**

**7.1.14 Method: is\_ampel**

**7.1.15 Method: iox**

## Chapter 8

# ServiceCleware.py

### 8.1 Class: ServiceCleware

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.Cleware  
↪ import ServiceCleware
```

Service for controlling Cleware devices.

This class extends ServiceBase to provide functionalities specific to managing and controlling Cleware devices.

#### 8.1.1 Method: svc\_api\_get\_all\_devices\_state

Retrieve the state of all Cleware devices.

**Returns:**

*/ Type: dict /*

A dictionary containing the states of all Cleware devices.

#### 8.1.2 Method: svc\_api\_set\_switch

Set state for a Cleware device's switch.

**Arguments:**

- device\_no  
*/ Condition: required / Type: str /*  
Cleware device's number.
- switch\_id  
*/ Condition: required / Type: str /*  
Switch number to turn on or off.
- state  
*/ Condition: required / Type: str /*  
State of switch to set (on/off).

**Returns:**

*/ Type: int /*

Return ret code, 1 for succeed, 0 for failure.

### 8.1.3 Method: `notify_updates`

Notify updates to the realtime update channel for Cleware devices.

**Returns:**

*(no returns)*

## Chapter 9

# ServiceLogger.py

### 9.1 Class: ServiceLogger

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.Clewa  
↳ import ServiceLogger
```

Logger class. microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-clewareservice-  
servicelogger-servicelogger-set-logger -----  
-----

#### 9.1.1 Method: get\_config\_file

Get a name of a caller in the format module.class.method Args: skip: specifies how many levels of stack to skip while getting caller name. skip=1 means "who calls me", skip=2 "who calls my caller" etc.

**Returns:** An empty string is returned if skipped levels exceed stack height

#### 9.1.2 Method: log

#### 9.1.3 Method: log\_info

#### 9.1.4 Method: log\_debug

#### 9.1.5 Method: log\_warning

#### 9.1.6 Method: log\_error

#### 9.1.7 Method: log\_critical

# Chapter 10

## Utils.py

### 10.1 Class: Singleton

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.Cleware
↳ import Singleton
```

Class to implement Singleton Design Pattern. This class is used to derive the TTFisClientReal as only a single instance of this class is allowed.

Disabled pyLint Messages: R0903: Too few public methods (%s/%s) Used when class has too few public methods, so be sure it's really worth it.

This base class implements the Singleton Design Pattern required for the TTFisClientReal. Adding further methods does not make sense.

### 10.2 Class: Utils

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.Cleware
↳ import Utils
```

Class to implement utilities for supporting development. microservicebase-microservicemanagergui-dist-win-unpacked-resources-python-services-cleware-service-utils-utils-get-all-descendant-classes -----

Get all descendant classes of a class Args: cls: Input class for finding descendants.

**Returns:** Array of descendant classes.

#### 10.2.1 Method: get\_all\_sub\_classes

Get all children classes of a class Args: cls: Input class for finding children.

**Returns:** Array of children classes.

#### 10.2.2 Method: caller\_name

Get a name of a caller in the format module.class.method Args: skip: specifies how many levels of stack to skip while getting caller name. skip=1 means "who calls me", skip=2 "who calls my caller" etc.

**Returns:** An empty string is returned if skipped levels exceed stack height

### 10.2.3 Method: load\_library

Load native library depend on the calling convention. Args: path: library path. is\_stdcall: determine if the library's calling convention is stdcall or cdecl.

**Returns:** Loaded library object.

### 10.2.4 Method: is\_ascii\_or\_unicode

Check if the string is ascii or unicode Args: str\_check: string for checking codecs: encoding type list

**Returns:** True : if checked string is ascii or unicode False : if checked string is not ascii or unicode

### 10.2.5 Method: get\_registry\_parameters

Get service's parameters which stored in registry. Args: service\_name: Name of the service. key\_name: Name of register key.

**Returns:** Value of service's parameters

### 10.2.6 Method: create\_registry\_parameters

Create service's parameters and store it to registry. Args: service\_name: Name of the service. key\_name: Name of register key. parameters: Parameters to be stored.

**Returns:** None

## 10.3 Class: Job

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.dist.win-unpacked.resources.python.services.Clew
↳ import Job
```

### 10.3.1 Method: stop

### 10.3.2 Method: run

## Chapter 11

### `__main__.py`

11.1 Function: `main`

11.2 Function: `signal_handler`

## Chapter 12

### main.py

12.1 Function: `main`

12.2 Function: `signal_handler`

## Chapter 13

# start\_bridge.py

Standalone FastAPI UI Bridge launcher.

Starts the FastAPI bridge that exposes microservices via REST API and WebSocket. Can be spawned by the Electron GUI or run directly from the command line.

### 13.1 Function: `parse_args`

### 13.2 Function: `main`

## Chapter 14

# start\_registry.py

Standalone Service Registry launcher.

Starts the Service Registry that tracks all services, handles registration events, and broadcasts updates to UI clients via a fanout exchange.

### 14.1 Function: `parse_args`

### 14.2 Function: `main`

## Chapter 15

### `__init__.py`

biplist -- a library for reading and writing binary property list files.

Binary Property List (plist) files provide a faster and smaller serialization format for property lists on OS X. This is a library for generating binary plists which can be read by OS X, iOS, or other clients.

The API models the plistlib API, and will call through to plistlib when XML serialization or deserialization is required.

To generate plists with UID values, wrap the values with the `Uid` object. The value must be an int.

To generate plists with `NSData`/`CFData` values, wrap the values with the `Data` object. The value must be a string.

Date values can only be `datetime.datetime` objects.

The exceptions `InvalidPlistException` and `NotBinaryPlistException` may be thrown to indicate that the data cannot be serialized or deserialized as a binary plist.

Plist generation example:

```
from biplist import * from datetime import datetime plist = {'aKey':'aValue', '0':1.322, 'now':datetime.now(),
'list':[1,2,3], 'tuple':('a','b','c')} try: writePlist(plist, "example.plist") except (InvalidPlistException,
NotBinaryPlistException), e: print "Something bad happened:", e
```

Plist parsing example:

```
from biplist import * try: plist = readPlist("example.plist") print plist except (InvalidPlistException,
NotBinaryPlistException), e: print "Not a plist:", e
```

#### 15.1 Function: `readPlist`

Raises `NotBinaryPlistException`, `InvalidPlistException` `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---wrapdataobject =====`

#### 15.2 Function: `writePlist`

#### 15.3 Function: `readPlistFromString`

#### 15.4 Function: `writePlistToString`

#### 15.5 Function: `is_stream_binary_plist`

#### 15.6 Class: `Uid`

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import Uid

```

Wrapper around integers for representing UID values. This is used in keyed archiving. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---data =====

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import Data

```

Wrapper around bytes to distinguish Data values. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---invalidplistexception =====

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import InvalidPlistException

```

Raised when the plist is incorrectly formatted. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---notbinaryplistexception =====

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import NotBinaryPlistException

```

Raised when a binary plist was expected but not encountered. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistreader =====

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import PlistReader

```

- 15.6.1 Method: parse
- 15.6.2 Method: reset
- 15.6.3 Method: readRoot
- 15.6.4 Method: setCurrentOffsetToObjectNumber
- 15.6.5 Method: beginOffsetProtection
- 15.6.6 Method: endOffsetProtection
- 15.6.7 Method: readObject
- 15.6.8 Method: readContents
- 15.6.9 Method: readInteger
- 15.6.10 Method: readReal
- 15.6.11 Method: readRefs
- 15.6.12 Method: readArray
- 15.6.13 Method: readDict
- 15.6.14 Method: readAsciiString
- 15.6.15 Method: readUnicode
- 15.6.16 Method: readDate
- 15.6.17 Method: readData
- 15.6.18 Method: readUid
- 15.6.19 Method: getSizedInteger

Numbers of 8 bytes are signed integers when they refer to numbers, but unsigned otherwise. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---hashablewrapper =====

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import HashableWrapper
```

## 15.7 Class: BoolWrapper

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import BoolWrapper
```

## 15.8 Class: FloatWrapper

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import FloatWrapper
```

## 15.9 Class: StringWrapper

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import StringWrapper
```

### 15.9.1 Method: encodingMarker

## 15.10 Class: PlistWriter

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.biplist.__init__
↳ import PlistWriter
```

### 15.10.1 Method: reset

### 15.10.2 Method: positionOfObjectReference

If the given object has been written already, return its position in the offset table. Otherwise, return None.  
microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeroot -----

Strategy is: - write header - wrap root object so everything is hashable - compute size of objects which will be written - need to do this in order to know how large the object refs will be in the list/dict/set reference lists - write objects - keep objects in writtenReferences - keep positions of object references in referencePositions - write object references with the length computed previously - computer object reference length - write object reference positions - write trailer microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-beginrecursionprotection -----

### 15.10.3 Method: endRecursionProtection

### 15.10.4 Method: wrapRoot

### 15.10.5 Method: incrementByteCount

### 15.10.6 Method: computeOffsets

### 15.10.7 Method: writeObjectReference

Tries to write an object reference, adding it to the references table. Does not write the actual object bytes or set the reference position. Returns a tuple of whether the object was a new reference (True if it was, False if it already was in the reference table) and the new output. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeobject -----

Serializes the given object to the output. Returns output. If setReferencePosition is True, will set the position the object was written. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-writeoffsettable -----

Writes all of the object reference offsets. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-binaryreal` -----  
-----

#### 15.10.8 Method: `binaryInt`

#### 15.10.9 Method: `intSize`

Returns the number of bytes necessary to store the given integer. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-biplist---init---plistwriter-realsize` -----  
-----

## Chapter 16

# badge.py

### 16.1 Function: `badge_disk_icon`

# Chapter 17

## colors.py

### 17.1 Function: isAColor

### 17.2 Function: parseColor

### 17.3 Class: Color

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import Color
```

#### 17.3.1 Method: to\_rgb

### 17.4 Class: RGB

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import RGB
```

#### 17.4.1 Method: to\_rgb

### 17.5 Class: HSL

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import HSL
```

#### 17.5.1 Method: to\_rgb

### 17.6 Class: HWB

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import HWB
```

### 17.6.1 Method: to\_rgb

## 17.7 Class: CMYK

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import CMYK
```

### 17.7.1 Method: to\_rgb

## 17.8 Class: Gray

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import Gray
```

### 17.8.1 Method: to\_rgb

## 17.9 Class: ColorParser

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.colors
↳ import ColorParser
```

- 17.9.1 Method: skipws
- 17.9.2 Method: expect
- 17.9.3 Method: expectEnd
- 17.9.4 Method: getToken
- 17.9.5 Method: parseNumber
- 17.9.6 Method: parseColor
- 17.9.7 Method: parseRGB
- 17.9.8 Method: parseHSL
- 17.9.9 Method: parseHWB
- 17.9.10 Method: parseCMYK
- 17.9.11 Method: parseGray
- 17.9.12 Method: parseValue
- 17.9.13 Method: parseAngle

# Chapter 18

## core.py

### 18.1 Function: build\_dmg

### 18.2 Class: DMGError

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.dmgbuild.core  
↳ import DMGError
```

# Chapter 19

## buddy.py

### 19.1 Class: BuddyError

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.buddy  
↪ import BuddyError
```

### 19.2 Class: Block

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.buddy  
↪ import Block
```

**19.2.1 Method: close**

**19.2.2 Method: flush**

**19.2.3 Method: invalidate**

**19.2.4 Method: zero\_fill**

**19.2.5 Method: tell**

**19.2.6 Method: seek**

**19.2.7 Method: read**

**19.2.8 Method: write**

**19.2.9 Method: insert**

**19.2.10 Method: delete**

### 19.3 Class: Allocator

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds-store.buddy
↳ import Allocator

```

### 19.3.1 Method: open

### 19.3.2 Method: close

### 19.3.3 Method: flush

### 19.3.4 Method: read

Read data at 'offset', or raise an exception. 'size\_or\_format' may either be a byte count, in which case we return raw data, or a format string for 'struct.unpack', in which case we work out the size and unpack the data before returning it. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-write -----

Write data at 'offset', or raise an exception. 'data\_or\_format' may either be the data to write, or a format string for 'struct.pack', in which case we pack the additional arguments and write the resulting data. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-get-block -----

### 19.3.5 Method: allocate

Allocate or reallocate a block such that it has space for at least 'bytes' bytes. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-buddy-allocator-release -----

### 19.3.6 Method: iterkeys

### 19.3.7 Method: keys

# Chapter 20

## store.py

### 20.1 Class: ILocCodec

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store  
↳ import ILocCodec
```

#### 20.1.1 Method: encode

#### 20.1.2 Method: decode

### 20.2 Class: PlistCodec

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store  
↳ import PlistCodec
```

#### 20.2.1 Method: encode

#### 20.2.2 Method: decode

### 20.3 Class: BookmarkCodec

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store  
↳ import BookmarkCodec
```

#### 20.3.1 Method: encode

#### 20.3.2 Method: decode

### 20.4 Class: DSStoreEntry

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import DSStoreEntry

```

Holds the data from an entry in a `.DS_Store` file. Note that this is not meant to represent the entry itself--i.e. if you change the type or value, your changes will *not* be reflected in the underlying file.

If you want to make a change, you should either use the `DSStore` object's `DSStore.insert` method (which will replace a key if it already exists), or the mapping access mode for `DSStore` (often simpler anyway). `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-read` -----

Read a `.DS_Store` entry from the containing Block `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-byte-length` -----

Compute the length of this entry, in bytes `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstoreentry-write` -----

Write this entry to the specified Block `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore` =====

*Imported by:*

```

from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.ds_store.store
↳ import DSStore

```

Python interface to a `.DS_Store` file. Works by manipulating the file on the disk--so this code will work with `.DS_Store` files for *very* large directories.

A `DSStore` object can be used as if it was a mapping, e.g.:

```
d['foobar.dat']['Iloc']
```

will fetch the "Iloc" record for "foobar.dat", or raise `KeyError` if there is no such record. If used in this manner, the `DSStore` object will return (type, value) tuples, unless the type is "blob" and the module knows how to decode it.

Currently, we know how to decode "Iloc", "bwsp", "lsvp", "lsvP" and "icvp" blobs. "Iloc" decodes to an (x, y) tuple, while the others are all decoded using `biplist`.

Assignment also works, e.g.:

```
d['foobar.dat']['note'] = ('ustr', u'Hello World!')
```

as does deletion with `del`:

```
del d['foobar.dat']['note']
```

This is usually going to be the most convenient interface, though occasionally (for instance when creating a new `.DS_Store` file) you may wish to drop down to using `DSStoreEntry` objects directly. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-open` -----

Open a `.DS_Store` file; pass either a Python file object, or a filename in the `file_or_name` argument and a file access mode in the `mode` argument. If you are creating a new file using the "w" or "w+" modes, you may also specify a list of entries with which to initialise the file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-flush` -----

Flush any dirty data back to the file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-close` -----

Flush dirty data and close the underlying file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-insert` -----

Insert entry (which should be a `DSStoreEntry`) into the B-Tree. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-delete` -----

Delete an item, identified by `filename` and `code` from the B-Tree. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-ds-store-store-dsstore-find` -----  
-----

Returns a generator that will iterate over matching entries in the B-Tree.

# Chapter 21

## alias.py

### 21.1 Function: encode\_utf8

### 21.2 Function: decode\_utf8

### 21.3 Class: AppleShareInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import AppleShareInfo
```

### 21.4 Class: VolumeInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import VolumeInfo
```

#### 21.4.1 Method: filesystem\_type

### 21.5 Class: TargetInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import TargetInfo
```

### 21.6 Class: Alias

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.alias
↳ import Alias
```

### 21.6.1 Method: from\_bytes

Construct an `Alias` object given binary `Alias` data. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-alias-alias-for-file -----`  
-----

Create an `Alias` that points at the specified file. `microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-alias-alias-to-bytes -----`  
-----

Returns the binary representation for this `Alias`.

## Chapter 22

# bookmark.py

### 22.1 Function: iteritems

### 22.2 Class: Data

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import Data
```

### 22.3 Class: URL

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import URL
```

#### 22.3.1 Method: absolute

Return an absolute URL. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark =====

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.bookmar
↳ import Bookmark
```

#### 22.3.2 Method: from\_bytes

Create a `Bookmark` given byte data. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-get -----

Lookup the value for a given key, returning a default if not present. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-to-bytes -----

Convert this `Bookmark` to a byte representation. microservicebase-microservicemanagergui-node-modules-dmg-builder-vendor-mac-alias-bookmark-bookmark-for-file -----

Construct a `Bookmark` for a given file.

# Chapter 23

## osx.py

### 23.1 Function: getattrlist

### 23.2 Function: fgetattrlist

### 23.3 Function: statfs

### 23.4 Function: fstatfs

### 23.5 Class: attrlist

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attrlist
```

### 23.6 Class: attribute\_set\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attribute_set_t
```

### 23.7 Class: fsobj\_id\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import fsobj_id_t
```

### 23.8 Class: timespec

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import timespec
```

## 23.9 Class: attrreference\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import attrreference_t
```

## 23.10 Class: fsid\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import fsid_t
```

## 23.11 Class: guid\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import guid_t
```

## 23.12 Class: kauth\_ace

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_ace
```

## 23.13 Class: kauth\_acl

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_acl
```

## 23.14 Class: kauth\_filesec

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import kauth_filesec
```

## 23.15 Class: diskextent

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import diskextent
```

## 23.16 Class: Point

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import Point
```

## 23.17 Class: Rect

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import Rect
```

## 23.18 Class: FileInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FileInfo
```

## 23.19 Class: FolderInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FolderInfo
```

## 23.20 Class: FinderInfo

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import FinderInfo
```

## 23.21 Class: vol\_capabilities\_attr\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import vol_capabilities_attr_t
```

## 23.22 Class: vol\_attributes\_attr\_t

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import vol_attributes_attr_t
```

## 23.23 Class: struct\_statfs

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.osx
↳ import struct_statfs
```

# Chapter 24

## utils.py

### 24.1 Class: UTC

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.node_modules.dmg-builder.vendor.mac_alias.utils  
↪ import UTC
```

#### 24.1.1 Method: utcoffset

#### 24.1.2 Method: dst

#### 24.1.3 Method: tzname

## Chapter 25

# launcher.py

FastAPI Bridge launcher.

Starts the FastAPI Bridge (REST/WS gateway) that connects the MicroserviceManagerGUI to microservices via RabbitMQ.

### 25.1 Function: `parse_args`

### 25.2 Function: `main`

## Chapter 26

# ClewareAccessHelper.py

### 26.1 Class: ClewareAccessHelper

*Imported by:*

```
from
↳ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ClewareAccessHelper
↳ import ClewareAccessHelper
```

ClewareAccessHelper class acts as a proxy class in Proxy pattern and forward the request to the real helper depend on platform. microservicebase-microservicemanagergui-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-import-modules-from-paths -----

Import all modules from given paths.

**Arguments:**

- paths  
/ Condition: required / Type: list /  
List of paths to import modules from.

#### 26.1.1 Method: load\_config

Load the user config port name for USB access. Returns: None microservicebase-microservicemanagergui-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-get-config -----

Get Cleware ports' config. Returns: Dictionary of ports' setting. microservicebase-microservicemanagergui-python-services-clewareservice-clewareaccesshelper-clewareaccesshelper-set-config -----

Save Cleware ports' config to file. Args: config\_json: data to be saved.

**Returns:** res: saved result.

- 26.1.2 Method: `init_cleware`
- 26.1.3 Method: `open_cleware`
- 26.1.4 Method: `close_cleware`
- 26.1.5 Method: `get_handle`
- 26.1.6 Method: `set_value`
- 26.1.7 Method: `get_value`
- 26.1.8 Method: `set_switch`
- 26.1.9 Method: `set_switch_by_port_name`
- 26.1.10 Method: `get_switch`
- 26.1.11 Method: `get_version`
- 26.1.12 Method: `get_usb_type`
- 26.1.13 Method: `get_serial_number`
- 26.1.14 Method: `valid_ser_number`
- 26.1.15 Method: `get_hw_version`
- 26.1.16 Method: `is_ampel`
- 26.1.17 Method: `iox`
- 26.1.18 Method: `get_all_devices_state`

## Chapter 27

# ClewareAccessHelperAbs.py

### 27.1 Class: ClewareAccessHelperAbs

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ClewareAccessHelperAbs  
↳ import ClewareAccessHelperAbs
```

ClewareAccessHelperAbs acts as a abstract class for the real Helper in Proxy Pattern.    microservicebase-microservicemanagergui-python-services-clewareservice-clewareaccesshelperabs-clewareaccesshelperabs-init-cleware

-----

**27.1.1 Method: open\_cleware**

**27.1.2 Method: close\_cleware**

**27.1.3 Method: set\_value**

**27.1.4 Method: get\_value**

**27.1.5 Method: set\_switch**

**27.1.6 Method: get\_switch**

**27.1.7 Method: get\_handle**

**27.1.8 Method: get\_version**

**27.1.9 Method: get\_usb\_type**

**27.1.10 Method: get\_usb\_type**

**27.1.11 Method: get\_serial\_number**

**27.1.12 Method: valid\_ser\_number**

**27.1.13 Method: get\_hw\_version**

**27.1.14 Method: is\_ampel**

**27.1.15 Method: iox**

## Chapter 28

# ClewareAccessHelperLinux.py

### 28.1 Class: ClewareAccessHelperLinux

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ClewareAccessHelp  
↪ import ClewareAccessHelperLinux
```

**ClewareAccessHelperLinux acts as the real object on Linux platform in Proxy Pattern**

- This is the wrapper class for the functions provided by C/C++ source for Linux.
- The C/C++ source for Linux is provided by Cleware manufacturer at : [http://www.cleware-shop.de/Downloads\\_EN](http://www.cleware-shop.de/Downloads_EN)
- The C/C++ source for Linux is modified by Cuong Nguyen (RBVH/ENG22) for support python to load dynamic library on Linux.

- 28.1.1 Method: `init_cleware`
- 28.1.2 Method: `open_cleware`
- 28.1.3 Method: `close_cleware`
- 28.1.4 Method: `get_handle`
- 28.1.5 Method: `set_value`
- 28.1.6 Method: `get_value`
- 28.1.7 Method: `set_switch`
- 28.1.8 Method: `get_switch`
- 28.1.9 Method: `get_version`
- 28.1.10 Method: `get_usb_type`
- 28.1.11 Method: `get_serial_number`
- 28.1.12 Method: `valid_ser_number`
- 28.1.13 Method: `get_hw_version`
- 28.1.14 Method: `is_ampel`
- 28.1.15 Method: `iox`
- 28.1.16 Method: `get_all_sw_state`

## Chapter 29

# ClewareAccessHelperWindows.py

### 29.1 Class: ClewareAccessHelperWindows

*Imported by:*

```
from  
↳ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ClewareAccessHelp  
↳ import ClewareAccessHelperWindows
```

**ClewareAccessHelperWindows acts as the real object on Windows platform in Proxy Pattern**

- This is the wrapper class for the functions provided by USBaccess.dll.
- The USBaccess.dll is provided by Cleware manufacturer at : [http://www.cleware-shop.de/Downloads\\_EN](http://www.cleware-shop.de/Downloads_EN)

**29.1.1 Method: init\_cleware**

**29.1.2 Method: open\_cleware**

**29.1.3 Method: close\_cleware**

**29.1.4 Method: get\_handle**

**29.1.5 Method: set\_value**

**29.1.6 Method: get\_value**

**29.1.7 Method: set\_switch**

**29.1.8 Method: get\_switch**

**29.1.9 Method: get\_version**

**29.1.10 Method: get\_usb\_type**

**29.1.11 Method: get\_serial\_number**

**29.1.12 Method: valid\_ser\_number**

**29.1.13 Method: get\_hw\_version**

**29.1.14 Method: is\_ampel**

**29.1.15 Method: iox**

## Chapter 30

# ServiceCleware.py

### 30.1 Class: ServiceCleware

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ServiceCleware  
↪ import ServiceCleware
```

Service for controlling Cleware devices.

This class extends ServiceBase to provide functionalities specific to managing and controlling Cleware devices.

#### 30.1.1 Method: svc\_api\_get\_all\_devices\_state

Retrieve the state of all Cleware devices.

**Returns:**

*/ Type: dict /*

A dictionary containing the states of all Cleware devices.

#### 30.1.2 Method: svc\_api\_set\_switch

Set state for a Cleware device's switch.

**Arguments:**

- device\_no  
*/ Condition: required / Type: str /*  
Cleware device's number.
- switch\_id  
*/ Condition: required / Type: str /*  
Switch number to turn on or off.
- state  
*/ Condition: required / Type: str /*  
State of switch to set (on/off).

**Returns:**

*/ Type: int /*

Return ret code, 1 for succeed, 0 for failure.

### 30.1.3 Method: `notify_updates`

Notify updates to the realtime update channel for Cleware devices.

**Returns:**

*(no returns)*

# Chapter 31

## ServiceLogger.py

### 31.1 Class: ServiceLogger

*Imported by:*

```
from  
↪ MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.ServiceLogger  
↪ import ServiceLogger
```

Logger class.    microservicebase-microservicemanagergui-python-services-clewareservice-servicelogger-servicelogger-set-logger -----

#### 31.1.1 Method: get\_config\_file

Get a name of a caller in the format module.class.method Args: skip: specifies how many levels of stack to skip while getting caller name. skip=1 means "who calls me", skip=2 "who calls my caller" etc.

**Returns:** An empty string is returned if skipped levels exceed stack height

#### 31.1.2 Method: log

#### 31.1.3 Method: log\_info

#### 31.1.4 Method: log\_debug

#### 31.1.5 Method: log\_warning

#### 31.1.6 Method: log\_error

#### 31.1.7 Method: log\_critical

# Chapter 32

## Utils.py

### 32.1 Class: Singleton

*Imported by:*

```
from MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.Utils
↳ import Singleton
```

Class to implement Singleton Design Pattern. This class is used to derive the TTFisClientReal as only a single instance of this class is allowed.

Disabled pyLint Messages: R0903: Too few public methods (%s/%s) Used when class has too few public methods, so be sure it's really worth it.

This base class implements the Singleton Design Pattern required for the TTFisClientReal. Adding further methods does not make sense.

### 32.2 Class: Utils

*Imported by:*

```
from MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.Utils
↳ import Utils
```

Class to implement utilities for supporting development. microservicebase-microservicemanagergui-python-services-clewareservice-utils-utils-get-all-descendant-classes -----  
-----

Get all descendant classes of a class Args: cls: Input class for finding descendants.

**Returns:** Array of descendant classes.

#### 32.2.1 Method: get\_all\_sub\_classes

Get all children classes of a class Args: cls: Input class for finding children.

**Returns:** Array of children classes.

#### 32.2.2 Method: caller\_name

Get a name of a caller in the format module.class.method Args: skip: specifies how many levels of stack to skip while getting caller name. skip=1 means "who calls me", skip=2 "who calls my caller" etc.

**Returns:** An empty string is returned if skipped levels exceed stack height

### 32.2.3 Method: load\_library

Load native library depend on the calling convention. Args: path: library path. is\_stdcall: determine if the library's calling convention is stdcall or cdecl.

**Returns:** Loaded library object.

### 32.2.4 Method: is\_ascii\_or\_unicode

Check if the string is ascii or unicode Args: str\_check: string for checking codecs: encoding type list

**Returns:** True : if checked string is ascii or unicode False : if checked string is not ascii or unicode

### 32.2.5 Method: get\_registry\_parameters

Get service's parameters which stored in registry. Args: service\_name: Name of the service. key\_name: Name of register key.

**Returns:** Value of service's parameters

### 32.2.6 Method: create\_registry\_parameters

Create service's parameters and store it to registry. Args: service\_name: Name of the service. key\_name: Name of register key. parameters: Parameters to be stored.

**Returns:** None

## 32.3 Class: Job

*Imported by:*

```
from MicroserviceBase.MicroserviceManagerGUI.python.services.ClewareService.Utils
↳ import Job
```

### 32.3.1 Method: stop

### 32.3.2 Method: run

## Chapter 33

### `__main__.py`

**33.1**    Function: `main`

**33.2**    Function: `signal_handler`

## Chapter 34

### main.py

34.1 Function: `main`

34.2 Function: `signal_handler`

## Chapter 35

# start\_bridge.py

Standalone FastAPI UI Bridge launcher.

Starts the FastAPI bridge that exposes microservices via REST API and WebSocket. Can be spawned by the Electron GUI or run directly from the command line.

### 35.1 Function: `parse_args`

### 35.2 Function: `main`

## Chapter 36

# start\_registry.py

Standalone Service Registry launcher.

Starts the Service Registry that tracks all services, handles registration events, and broadcasts updates to UI clients via a fanout exchange.

### 36.1 Function: `parse_args`

### 36.2 Function: `main`

# Chapter 37

## eventbus\_config.py

### 37.1 Class: EventBusConfig

*Imported by:*

```
from MicroserviceBase.adapters.config.eventbus_config import EventBusConfig
```

Configuration for EventBusClient-based adapters.

Wraps a JSONP config file path used by EventBusClient.from\_config\_sync().

#### 37.1.1 Method: load\_config

Load and return the parsed config as a dict.

Requires EventBusClient to be installed.

**Returns:**

*/ Type: dict /*

Parsed configuration dictionary.

#### 37.1.2 Method: to\_config\_source

Load config and return as a dict with optional field overrides.

Useful for creating derivative clients (e.g., registry or fanout) that share connection settings but differ in exchange configuration.

**Arguments:**

- overrides

*/ Condition: optional / Type: keyword arguments /*

Fields to override in the loaded config (e.g., exchange\_name, exchange\_handler).

**Returns:**

*/ Type: dict /*

Config dictionary with overrides applied.

## Chapter 38

# rabbitmq\_config.py

### 38.1 Class: RabbitMQConfig

*Imported by:*

```
from MicroserviceBase.adapters.config.rabbitmq.config import RabbitMQConfig
```

Configuration for connecting to RabbitMQ.

#### 38.1.1 Method: to\_connection\_params

Convert to pika ConnectionParameters kwargs.

**Returns:**

dict suitable for pika.ConnectionParameters(\*\*kwargs).

#### 38.1.2 Method: from\_cmd\_args

Parse RabbitMQ config from command-line arguments and environment variables.

**Arguments:**

- `cmd_args`  
/ *Condition*: optional / *Type*: list /  
List of CLI arguments, or None for sys.argv.
- `service_name`  
/ *Condition*: optional / *Type*: str /  
Name of the service (used in help text).

**Returns:**

RabbitMQConfig instance.

## Chapter 39

# local\_hub\_manager.py

Local Hub Manager — manages a local ProcessHub instance lifecycle.

Wraps ProcessHub APIs (lazy import) so the MicroserviceManager GUI can start/stop a ProcessHub on the local machine, manage processes, and optionally join a fleet as an agent.

### 39.1 Class: LocalHubManager

*Imported by:*

```
from MicroserviceBase.adapters.local_hub.local_hub_manager import LocalHubManager
```

Manages a local ProcessHub instance lifecycle. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-start-hub -----

Start a local ProcessHub server.

**Args:** mode: "standalone" or "agent" (join fleet). xpub\_port: ZMQ XPUB port for the broker. xsub\_port: ZMQ XSUB port for the broker. orchestrator\_url: Fleet orchestrator ZMQ address (agent mode). hub\_id: Hub identifier (agent mode). hub\_name: Human-readable hub name (agent mode). process\_config: Dict of {name: config} for processes.

**Returns:** Status dict.

#### 39.1.1 Method: stop\_hub

Stop the local hub. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-get-status -----

Get current hub status. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-start-processes ----

Start named processes. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-stop-processes ----

Stop named processes. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-get-config -----

Get all process configurations. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-add-config -----

Add a new process configuration. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-update-config -----

Update an existing process configuration. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-remove-config -----

Remove a process configuration. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-remove-service -----

Remove a service — delete config and managed service folder.

Only deletes the folder if it lives inside the managed <config\_dir>/services/ directory (never touches external paths). Stops the process first if it is still running. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-get-service-log -----

Read the last *tail* lines of a process log file. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-get-services-dir -----

Return absolute path to <config\_dir>/services/, creating it if needed. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-import-service -----

Import a microservice from a folder or base64-encoded ZIP.

Validates structure, copies files into <config\_dir>/services/{name}/, and creates a hub config entry using the \${python} placeholder.

Returns {"success": bool, "message": str, "warnings": [...]}. microservicebase-adapters-local-hub-local-hub-manager-localhubmanager-reset -----

Reset the hub (stop processes, clear connections).

## Chapter 40

# service\_executor.py

Service-aware process executor.

Wraps a `SimpleExecutor` and attempts graceful RPC shutdown (`svc_api_shutdown`) before falling back to signal-based stop.

### 40.1 Class: `ServiceExecutor`

*Imported by:*

```
from MicroserviceBase.adapters.local_hub.service_executor import ServiceExecutor
```

Process executor that sends an RPC shutdown to `MicroserviceBase` services.

Delegates all operations to a wrapped `SimpleExecutor`. On `stop()`, it first tries to send `svc_api_shutdown` via RabbitMQ to the service's queue. If the service exits within `shutdown_timeout` seconds the process is considered stopped. Otherwise it falls back to the delegate's signal-based stop (`CTRL_BREAK_EVENT` on Windows, `SIGTERM` on Linux). `microservicebase-adapters-local-hub-service-executor-serviceexecutor-start` -----  
-----

#### 40.1.1 Method: `get_log_path`

Public accessor for log file path. `microservicebase-adapters-local-hub-service-executor-serviceexecutor-read-log` -----  
-----

Public method to read process log. Returns the text content. `microservicebase-adapters-local-hub-service-executor-serviceexecutor-is-running` -----  
-----

#### 40.1.2 Method: `get_pid`

#### 40.1.3 Method: `stop`

# Chapter 41

## amqp\_registry\_adapter.py

### 41.1 Class: AMQPRegistryAdapter

*Imported by:*

```
from MicroserviceBase.adapters.registry.amqp-registry-adapter import  
↪ AMQPRegistryAdapter
```

AMQP implementation of the ServiceRegistryPort interface.

Uses RabbitMQ topic exchange for service registration events and fanout exchange for realtime update broadcasts.

#### 41.1.1 Method: register

Register a service by publishing to the service\_information exchange.

**Arguments:**

- service\_info  
/ *Condition*: required / *Type*: dict /  
dict containing service metadata.

#### 41.1.2 Method: unregister

Unregister a service by publishing to the service\_information exchange.

**Arguments:**

- service\_info  
/ *Condition*: required / *Type*: dict /  
dict containing service metadata.

#### 41.1.3 Method: subscribe\_to\_events

Subscribe to service registration events.

Runs in the calling thread (blocking). Typically called from a daemon thread.

**Arguments:**

- handler  
/ *Condition*: required / *Type*: callable /  
Callback function(ch, method, properties, body).

#### 41.1.4 Method: `notify_update`

Broadcast services update to the realtime update fanout exchange.

**Arguments:**

- `services_info`  
/ *Condition*: required / *Type*: dict /  
dict of all current service information.

#### 41.1.5 Method: `subscribe_to_updates`

Subscribe to the realtime update fanout exchange.

Unlike `subscribe_to_events` (which listens for individual on/off events), this subscribes to the full services dict broadcast by the Registry after each change. Uses an exclusive temporary queue to avoid competing consumers.

Runs in the calling thread (blocking). Typically called from a daemon thread.

**Arguments:**

- `handler`  
/ *Condition*: required / *Type*: callable /  
Callback function(`services_info_dict`) called with the full services dict.

#### 41.1.6 Method: `get_update_channel_name`

Return the realtime update exchange name.

#### 41.1.7 Method: `cleanup_update_exchange`

Delete the realtime update exchange (for cleanup on shutdown).

#### 41.1.8 Method: `cleanup`

Close event subscription resources and clean up the update exchange.

## Chapter 42

# eventbus\_registry\_adapter.py

### 42.1 Class: EventBusRegistryAdapter

*Imported by:*

```
from MicroserviceBase.adapters.registry.eventbus_registry_adapter import
↳ EventBusRegistryAdapter
```

EventBusClient implementation of the ServiceRegistryPort interface.

Uses EventBusClient with TopicExchangeHandler for service registration events and a FanoutExchangeHandler for realtime update broadcasts.

Both clients are created from the same JSONP config with exchange overrides via `EventBusConfig.to_config_source()`.

#### 42.1.1 Method: register

Register a service by publishing to the service\_information exchange.

**Arguments:**

- `service_info`  
/ *Condition:* required / *Type:* dict /  
dict containing service metadata.

#### 42.1.2 Method: unregister

Unregister a service by publishing to the service\_information exchange.

**Arguments:**

- `service_info`  
/ *Condition:* required / *Type:* dict /  
dict containing service metadata.

#### 42.1.3 Method: subscribe\_to\_events

Subscribe to service registration events.

Blocks the calling thread. Typically called from a daemon thread. The handler receives the pika-style callback signature (ch, method, properties, body) for backward compatibility.

**Arguments:**

- `handler`  
/ *Condition:* required / *Type:* callable /  
Callback function(ch, method, properties, body).

#### 42.1.4 Method: `notify_update`

Broadcast services update to the realtime update fanout exchange.

**Arguments:**

- `services_info`  
/ *Condition*: required / *Type*: dict /  
dict of all current service information.

#### 42.1.5 Method: `get_update_channel_name`

Return the realtime update exchange name.

#### 42.1.6 Method: `cleanup`

Close all EventBusClient connections.

## Chapter 43

# eventbus\_adapter.py

### 43.1 Class: `_ChannelProxy`

*Imported by:*

```
from MicroserviceBase.adapters.transport.eventbus_adapter import _ChannelProxy
```

Lightweight proxy that mimics a pika channel for the domain handler.

Translates `basic_publish` and `basic_ack` calls into `EventBusClient` operations.

#### 43.1.1 Method: `basic_publish`

Publish a reply message via the `EventBusClient`.

**Arguments:**

- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name (ignored, `EventBusClient` uses its configured exchange).
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for the reply (the `reply_to` value).
- `properties`  
/ *Condition*: optional / *Type*: object /  
Properties object with `correlation_id` attribute.
- `body`  
/ *Condition*: optional / *Type*: str or bytes /  
The message body (JSON string or bytes).

#### 43.1.2 Method: `basic_ack`

Acknowledge a message (no-op for `EventBusClient` which auto-acks).

### 43.2 Class: `_MethodProxy`

*Imported by:*

```
from MicroserviceBase.adapters.transport.eventbus_adapter import _MethodProxy
```

Lightweight proxy that mimics a pika method frame.

## 43.3 Class: \_PropertiesProxy

Imported by:

```
from MicroserviceBase.adapters.transport.eventbus_adapter import _PropertiesProxy
```

Lightweight proxy that mimics pika BasicProperties.

## 43.4 Class: EventBusTransportAdapter

Imported by:

```
from MicroserviceBase.adapters.transport.eventbus_adapter import  
↳ EventBusTransportAdapter
```

EventBusClient implementation of the TransportPort interface.

Creates the underlying EventBusClient from a JSONP configuration file via `EventBusClient.from_config_sync()`.

### 43.4.1 Method: connect

Create the EventBusClient from the JSONP config and establish connection.

### 43.4.2 Method: disconnect

Close the EventBusClient connection.

### 43.4.3 Method: consume

Start consuming RPC requests for a service.

Subscribes to the routing key and blocks the calling thread. Incoming messages are adapted to the pika-style handler signature.

**Arguments:**

- `service_name`  
/ *Condition*: required / *Type*: str /  
Name used as the service identifier.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for message binding.
- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name (must match the configured exchange).
- `handler`  
/ *Condition*: required / *Type*: callable /  
Callback function(ch, method, props, body).

### 43.4.4 Method: rpc\_call

Send an RPC request and wait for the response.

Creates a temporary subscription for the reply, sends the request with `reply_to` and `correlation_id` headers, and waits for the response.

**Arguments:**

- `request_data`  
/ *Condition*: required / *Type*: dict /  
dict to send as the request body.
- `exchange_name`  
/ *Condition*: required / *Type*: str /  
Exchange to publish to (must match configured exchange).
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for the target service.

**Returns:**

Parsed response dict.

### 43.4.5 Method: `publish`

Publish a message to the exchange.

**Arguments:**

- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name (must match configured exchange).
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key.
- `body`  
/ *Condition*: required / *Type*: str, bytes, or dict /  
Message body.
- `properties`  
/ *Condition*: optional / *Type*: dict /  
Optional message properties (used for headers).

## Chapter 44

# rabbitmq\_adapter.py

### 44.1 Class: RabbitMQTransportAdapter

*Imported by:*

```
from MicroserviceBase.adapters.transport.rabbitmq_adapter import  
↳ RabbitMQTransportAdapter
```

RabbitMQ implementation of the TransportPort interface.

#### 44.1.1 Method: connect

Establish connection to RabbitMQ.

#### 44.1.2 Method: disconnect

Close connection to RabbitMQ.

#### 44.1.3 Method: connection

Access the underlying pika connection (for backward compat).

#### 44.1.4 Method: stop\_consuming

Signal the consume loop to exit.

#### 44.1.5 Method: consume

Start consuming RPC requests for a service.

Sets up the exchange, queue, binding, and starts blocking consumption.

**Arguments:**

- `service_name`  
/ *Condition*: required / *Type*: str /  
Name used as the queue name.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for queue binding.

- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name to bind to.
- `handler`  
/ *Condition*: required / *Type*: callable /  
Callback function(ch, method, props, body).

#### 44.1.6 Method: `rpc_call`

Send an RPC request and wait for the response.

Creates a new connection for the RPC call (matching original behavior).

##### Arguments:

- `request_data`  
/ *Condition*: required / *Type*: dict /  
dict to serialize as the request body.
- `exchange_name`  
/ *Condition*: required / *Type*: str /  
Exchange to publish to.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for the target service.
- `timeout`  
/ *Condition*: optional / *Type*: int / *Default*: 30 /  
Timeout in seconds for waiting for the response.

##### Returns:

Parsed response dict.

#### 44.1.7 Method: `publish`

Publish a message to an exchange.

##### Arguments:

- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key.
- `body`  
/ *Condition*: required / *Type*: str or bytes /  
Message body (str or bytes).
- `properties`  
/ *Condition*: optional / *Type*: pika.BasicProperties /  
Optional pika.BasicProperties.

# Chapter 45

## fastapi\_bridge.py

### 45.1 Class: FastAPIBridge

*Imported by:*

```
from MicroserviceBase.adapters.ui_bridge.fastapi_bridge import FastAPIBridge
```

FastAPI implementation of UIBridgePort.

Exposes a REST API that any UI client (Electron, browser, mobile) can call. Also provides a WebSocket endpoint for push-based service updates.

**Endpoints:** POST /api/request - Route a service request through the transport GET /api/services - Get current services info WS /ws/updates - Real-time service update stream

Requires `fastapi` and `uvicorn` (optional dependencies).

#### 45.1.1 Method: start

Start the FastAPI server in a background thread.

**Arguments:**

- `request_handler`  
/ *Condition:* required / *Type:* callable /  
Callable(request\_data, exchange, routing\_key) -> response dict.
- `services_info_provider`  
/ *Condition:* required / *Type:* callable /  
Callable() -> services info dict.

#### 45.1.2 Method: wait

Block the calling thread until the server thread exits.

Uses a loop with timeout so that KeyboardInterrupt can be caught on Windows.

#### 45.1.3 Method: stop

Stop the FastAPI server.

#### 45.1.4 Method: broadcast\_update

Push a services update to all connected WebSocket clients.

**Arguments:**

- `services_info`  
/ *Condition*: required / *Type*: dict /  
dict of all current service information.

# Chapter 46

## exceptions.py

### 46.1 Class: ServiceError

*Imported by:*

```
from MicroserviceBase.domain.exceptions import ServiceError
```

Base exception for service-related errors.

### 46.2 Class: TransportError

*Imported by:*

```
from MicroserviceBase.domain.exceptions import TransportError
```

Raised when a transport-level operation fails.

### 46.3 Class: ServiceNotFoundError

*Imported by:*

```
from MicroserviceBase.domain.exceptions import ServiceNotFoundError
```

Raised when a requested service is not found in the registry.

### 46.4 Class: MethodNotFoundError

*Imported by:*

```
from MicroserviceBase.domain.exceptions import MethodNotFoundError
```

Raised when a requested method is not found on a service.

# Chapter 47

## messages.py

### 47.1 Class: ResultType

*Imported by:*

```
from MicroserviceBase.domain.messages import ResultType
```

Result types for service responses.

### 47.2 Class: ServiceRequest

*Imported by:*

```
from MicroserviceBase.domain.messages import ServiceRequest
```

Structured request to a service method.

#### 47.2.1 Method: to\_dict

Converts this `ServiceRequest` instance to a dictionary.

**Returns:**

*/ Type: dict /*

Dictionary representation of this request.

#### 47.2.2 Method: to\_json

Converts this `ServiceRequest` instance to a JSON string.

**Returns:**

*/ Type: str /*

JSON string representation of this request.

#### 47.2.3 Method: from\_dict

Creates a `ServiceRequest` instance from a dictionary.

**Arguments:**

- data

*/ Condition: required / Type: dict /*

Dictionary containing request fields.

**Returns:**

*/ Type: ServiceRequest /*  
New instance populated from the dictionary.

**47.2.4 Method: from\_json**

Creates a `ServiceRequest` instance from a JSON string.

**Arguments:**

- `json_str`  
*/ Condition: required / Type: str /*  
JSON string containing request fields.

**Returns:**

*/ Type: ServiceRequest /*  
New instance populated from the JSON string.

**47.3 Class: ServiceResponse**

*Imported by:*

```
from MicroserviceBase.domain.messages import ServiceResponse
```

Structured response from a service method.

**47.3.1 Method: to\_dict**

Converts this `ServiceResponse` instance to an ordered dictionary.

**Returns:**

*/ Type: OrderedDict /*  
Sorted ordered dictionary representation of this response.

**47.3.2 Method: to\_json**

Converts this `ServiceResponse` instance to a JSON string.

**Returns:**

*/ Type: str /*  
JSON string representation of this response.

**47.3.3 Method: get\_json**

Backward-compatible alias for `to_json()`.

**Returns:**

*/ Type: str /*  
JSON string representation of this response.

### 47.3.4 Method: `get_data`

Backward-compatible alias for `result_data`.

**Returns:**

*/ Type: Any /*

The result data of this response.

### 47.3.5 Method: `from_dict`

Creates a `ServiceResponse` instance from a dictionary.

**Arguments:**

- `data`

*/ Condition: required / Type: dict /*

Dictionary containing response fields.

**Returns:**

*/ Type: ServiceResponse /*

New instance populated from the dictionary.

### 47.3.6 Method: `from_json`

Creates a `ServiceResponse` instance from a JSON string.

**Arguments:**

- `json_str`

*/ Condition: required / Type: str /*

JSON string containing response fields.

**Returns:**

*/ Type: ServiceResponse /*

New instance populated from the JSON string.

## 47.4 Class: `ServiceEvent`

*Imported by:*

```
from MicroserviceBase.domain.messages import ServiceEvent
```

Event emitted when service state changes.

### 47.4.1 Method: `to_dict`

Converts this `ServiceEvent` instance to a dictionary.

**Returns:**

*/ Type: dict /*

Dictionary representation of this event.

### 47.4.2 Method: to\_json

Converts this `ServiceEvent` instance to a JSON string.

**Returns:**

*/ Type: str /*  
JSON string representation of this event.

### 47.4.3 Method: from\_dict

Creates a `ServiceEvent` instance from a dictionary.

**Arguments:**

- `data`  
*/ Condition: required / Type: dict /*  
Dictionary containing event fields.

**Returns:**

*/ Type: ServiceEvent /*  
New instance populated from the dictionary.

### 47.4.4 Method: from\_json

Creates a `ServiceEvent` instance from a JSON string.

**Arguments:**

- `json_str`  
*/ Condition: required / Type: str /*  
JSON string containing event fields.

**Returns:**

*/ Type: ServiceEvent /*  
New instance populated from the JSON string.

# Chapter 48

## models.py

### 48.1 Class: MethodArgument

*Imported by:*

```
from MicroserviceBase.domain.models import MethodArgument
```

Describes a single argument of a service API method.

#### 48.1.1 Method: to\_dict

Converts this MethodArgument instance to a dictionary.

**Returns:**

*/ Type: dict /*

Dictionary representation of this argument.

#### 48.1.2 Method: from\_dict

Creates a MethodArgument instance from a dictionary.

**Arguments:**

- data

*/ Condition: required / Type: dict /*

Dictionary containing argument fields.

**Returns:**

*/ Type: MethodArgument /*

New instance populated from the dictionary.

### 48.2 Class: ServiceMethod

*Imported by:*

```
from MicroserviceBase.domain.models import ServiceMethod
```

Describes a single API method exposed by a service.

### 48.2.1 Method: to\_dict

Converts this `ServiceMethod` instance to a dictionary.

**Returns:**

*/ Type: dict /*

Dictionary representation of this method.

### 48.2.2 Method: from\_dict

Creates a `ServiceMethod` instance from a name and dictionary.

**Arguments:**

- name

*/ Condition: required / Type: str /*

The method name.

- data

*/ Condition: required / Type: dict /*

Dictionary containing method fields.

**Returns:**

*/ Type: ServiceMethod /*

New instance populated from the dictionary.

## 48.3 Class: ServiceInfo

*Imported by:*

```
from MicroserviceBase.domain.models import ServiceInfo
```

Describes a service and its metadata.

### 48.3.1 Method: to\_dict

Converts this `ServiceInfo` instance to a dictionary.

**Returns:**

*/ Type: dict /*

Dictionary representation of this service info.

### 48.3.2 Method: from\_dict

Creates a `ServiceInfo` instance from a dictionary.

**Arguments:**

- data

*/ Condition: required / Type: dict /*

Dictionary containing service info fields.

**Returns:**

*/ Type: ServiceInfo /*

New instance populated from the dictionary.

# Chapter 49

## service\_base.py

### 49.1 Class: ServiceBase

*Imported by:*

```
from MicroserviceBase.domain.service_base import ServiceBase
```

Pure domain base class for services.

All methods used to export APIs of a service must begin with the prefix 'svc\_api'. This class handles API discovery, docstring parsing, and request dispatch. Transport and registry interactions are delegated to injected ports.

#### 49.1.1 Method: get\_service\_info

Return the ServiceInfo dataclass for this service.

#### 49.1.2 Method: get\_service\_info\_dict

Return the service info as a plain dict (backward compat).

#### 49.1.3 Method: serve

Start serving requests via the transport port.

#### 49.1.4 Method: register\_service

Register this service via the registry port.

#### 49.1.5 Method: unregister\_service

Unregister this service via the registry port.

#### 49.1.6 Method: request\_service

Send an RPC request to another service via the transport port.

**Arguments:**

- request\_data  
/ *Condition*: required / *Type*: dict /  
Dict with 'method' and 'args' keys (backward compat format).

- `exchange_name`  
/ *Condition*: required / *Type*: str /  
The exchange to publish to.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
The routing key for the target service.

**Returns:**

/ *Type*: dict /  
The response dict from the target service.

**49.1.7 Method: close**

Close the service.

**49.1.8 Method: create\_request\_data**

Create request data for a given method name and arguments.

**49.1.9 Method: get\_svc\_api\_methods\_dict**

Retrieve all service API methods (methods starting with 'svc\_api').

**49.1.10 Method: get\_svc\_api\_methods\_info\_dict**

Retrieve information for all service APIs from their docstrings.

**49.1.11 Method: parse\_docstring**

Parse function's docstring to get arguments and return information.

**49.1.12 Method: svc\_api\_get\_version**

Get the service version.

**Returns:**

/ *Type*: str /  
Version of the service.

**49.1.13 Method: svc\_api\_get\_gui\_files**

Compress and return GUI files as bytes.

**49.1.14 Method: svc\_api\_get\_gui\_checksum**

Get an MD5 checksum of all GUI files.

Returns None when gui.support is disabled or the GUIs directory is missing.

**Returns:**

/ *Type*: str /  
MD5 hex-digest of the GUI files, or None.

#### 49.1.15 Method: `svc_api_shutdown`

Gracefully shut down this service.

#### 49.1.16 Method: `is_specific_request`

Check if the request is a specific request. Override in subclasses.

#### 49.1.17 Method: `on_specific_request`

Handle a specific request. Override in subclasses.

##### Arguments:

- `request`  
/ *Condition*: required / *Type*: str /  
The request method name.
- `body`  
/ *Condition*: required / *Type*: dict /  
The parsed request body dict.

##### Returns:

/ *Type*: `ServiceResponse` /  
A `ServiceResponse`, or `None` if not handled.

#### 49.1.18 Method: `dispatch_request`

Dispatch a parsed request body and return a `ServiceResponse`.

This is the core domain logic: given a request dict with 'method' and 'args', find the matching API method, call it, and return a structured response.

##### Arguments:

- `request_body`  
/ *Condition*: required / *Type*: dict /  
Dict with 'method' and 'args' keys.

##### Returns:

/ *Type*: `ServiceResponse` /  
`ServiceResponse` with result type and data.

#### 49.1.19 Method: `on_request`

Handle an incoming request (transport callback signature).

This method is called by the transport adapter. It delegates to `dispatch_request` for the actual domain logic, then uses the transport to send the response back.

##### Arguments:

- `ch`  
/ *Condition*: required / *Type*: object /  
Channel object from transport.

- `method`  
/ *Condition*: required / *Type*: object /  
Delivery method from transport.
- `props`  
/ *Condition*: required / *Type*: object /  
Message properties from transport.
- `body`  
/ *Condition*: required / *Type*: bytes /  
Raw message body (bytes or dict).

# Chapter 50

## service\_registry.py

### 50.1 Class: ServiceRegistry

*Imported by:*

```
from MicroserviceBase.domain.service-registry import ServiceRegistry
```

Pure domain registry that tracks services, handles aliases, and routes requests.

#### 50.1.1 Method: handle\_update

Handle a service registration/unregistration event.

**Arguments:**

- `service_information`  
/ *Condition*: required / *Type*: dict /  
Dict with 'info' and 'state' keys.

#### 50.1.2 Method: notify\_updates

Notify connected clients about service updates via the registry port.

#### 50.1.3 Method: svc\_api\_shutdown

Gracefully shut down the Service Registry. `microservicebase-domain-service-registry-serviceregistry-svc-api-get-services-info` -----

Retrieve information of all services connected to the broker.

**Returns:**

/ *Type*: dict /  
A dictionary containing information of all connected services.

#### 50.1.4 Method: svc\_api\_get\_realtime\_update\_exchange

Retrieve the exchange name of the realtime update exchange.

**Returns:**

/ *Type*: str /  
The name of the realtime update exchange.

### 50.1.5 Method: `svc_api_update_alias_conf`

Update the alias configuration information.

**Arguments:**

- `alias_string`  
/ *Condition*: required / *Type*: str /  
The alias configuration string to be updated.

**Returns:**

(no returns)

### 50.1.6 Method: `svc_api_get_alias_conf`

Retrieve the alias configuration string in JSON format.

**Returns:**

/ *Type*: str /  
The alias configuration string in JSON format.

### 50.1.7 Method: `is_specific_request`

Check if the request is an alias-routed request.

### 50.1.8 Method: `handle_alias_request`

Handle an alias request by routing to the actual service.

**Arguments:**

- `body`  
/ *Condition*: required / *Type*: dict /  
Dict with 'method' and 'args' keys.

**Returns:**

/ *Type*: ServiceResponse /  
ServiceResponse or raw response dict from the target service.

# Chapter 51

## factory.py

### 51.1 Function: create\_transport

Create a TransportPort implementation.

**Arguments:**

- `transport_type`  
/ *Condition*: optional / *Type*: str /  
Transport backend identifier. Supports 'rabbitmq' and 'eventbus'.
- `cmd_args`  
/ *Condition*: optional / *Type*: list /  
Command-line arguments for config parsing (used by 'rabbitmq').
- `service_name`  
/ *Condition*: optional / *Type*: str /  
Service name for help text (used by 'rabbitmq').
- `config_path`  
/ *Condition*: optional / *Type*: str /  
Path to JSONP config file (used by 'eventbus').

**Returns:**

/ *Type*: TransportPort /  
A connected TransportPort instance.

### 51.2 Function: create\_registry

Create a ServiceRegistryPort implementation.

**Arguments:**

- `transport_type`  
/ *Condition*: optional / *Type*: str /  
Transport backend identifier. Supports 'rabbitmq' and 'eventbus'.
- `cmd_args`  
/ *Condition*: optional / *Type*: list /  
Command-line arguments for config parsing (used by 'rabbitmq').

- `service_name`  
/ *Condition*: optional / *Type*: str /  
Service name for help text (used by 'rabbitmq').
- `update_exchange_name`  
/ *Condition*: optional / *Type*: str /  
Name for the realtime update fanout exchange.
- `config_path`  
/ *Condition*: optional / *Type*: str /  
Path to JSONP config file (used by 'eventbus').

**Returns:**

/ *Type*: ServiceRegistryPort /  
A ServiceRegistryPort instance.

## 51.3 Function: create\_ui\_bridge

Create a UIBridgePort implementation.

**Arguments:**

- `bridge_type`  
/ *Condition*: optional / *Type*: str /  
Bridge backend identifier. Currently supports 'fastapi'.
- `host`  
/ *Condition*: optional / *Type*: str /  
Host to bind to.
- `port`  
/ *Condition*: optional / *Type*: int /  
Port to bind to.

**Returns:**

/ *Type*: UIBridgePort /  
A UIBridgePort instance (not yet started).

# Chapter 52

## registry.py

### 52.1 Class: ServiceRegistryPort

*Imported by:*

```
from MicroserviceBase.ports.registry import ServiceRegistryPort
```

Abstract interface for service registration and discovery.

Implementations handle how services announce themselves and how the registry discovers/tracks them.

#### 52.1.1 Method: register

Register a service with the registry.

**Arguments:**

- `service_info`  
/ *Condition*: required / *Type*: dict /  
Dict containing service metadata.

#### 52.1.2 Method: unregister

Unregister a service from the registry.

**Arguments:**

- `service_info`  
/ *Condition*: required / *Type*: dict /  
Dict containing service metadata.

#### 52.1.3 Method: subscribe\_to\_events

Subscribe to service registration/unregistration events.

**Arguments:**

- `handler`  
/ *Condition*: required / *Type*: callable /  
Callback function(ch, method, properties, body) for events.

#### 52.1.4 Method: `notify_update`

Broadcast a services update to all subscribers.

**Arguments:**

- `services_info`  
/ *Condition*: required / *Type*: dict /  
Dict of all current service information.

#### 52.1.5 Method: `get_update_channel_name`

Return the name of the realtime update channel/exchange.

**Returns:**

Channel/exchange name as a string.

# Chapter 53

## transport.py

### 53.1 Class: TransportPort

*Imported by:*

```
from MicroserviceBase.ports.transport import TransportPort
```

Abstract interface for message transport.

Implementations provide the actual connectivity (e.g., RabbitMQ, Redis, MQTT). The domain layer depends only on this interface.

#### 53.1.1 Method: connect

Establish connection to the message broker.

#### 53.1.2 Method: disconnect

Close connection to the message broker.

#### 53.1.3 Method: stop\_consuming

Signal the consume loop to stop.

#### 53.1.4 Method: consume

Start consuming messages for a service.

**Arguments:**

- `service_name`  
/ *Condition*: required / *Type*: str /  
Name of the service (used as queue name).
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key to bind the queue.
- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name to bind to.
- `handler`  
/ *Condition*: required / *Type*: callable /  
Callback function(ch, method, props, body) for incoming messages.

### 53.1.5 Method: `rpc_call`

Send an RPC request and wait for the response.

**Arguments:**

- `request_data`  
/ *Condition*: required / *Type*: dict /  
Dict to serialize and send as the request body.
- `exchange_name`  
/ *Condition*: required / *Type*: str /  
Exchange to publish to.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key for the target service.

**Returns:**

Parsed response dict from the target service.

### 53.1.6 Method: `publish`

Publish a message to an exchange.

**Arguments:**

- `exchange`  
/ *Condition*: required / *Type*: str /  
Exchange name.
- `routing_key`  
/ *Condition*: required / *Type*: str /  
Routing key.
- `body`  
/ *Condition*: required / *Type*: str or bytes /  
Message body.
- `properties`  
/ *Condition*: optional / *Type*: dict /  
Optional message properties dict.

# Chapter 54

## ui\_bridge.py

### 54.1 Class: UIBridgePort

*Imported by:*

```
from MicroserviceBase.ports.ui_bridge import UIBridgePort
```

Abstract interface for UI communication bridge.

Implementations provide the actual UI connectivity (e.g., FastAPI REST, gRPC). Any frontend (Electron, browser, mobile) can connect through this port.

#### 54.1.1 Method: start

Start the UI bridge server.

**Arguments:**

- `request_handler`  
/ *Condition*: required / *Type*: callable /  
Callable that accepts a `ServiceRequest` dict and returns a `ServiceResponse` dict. Used to route UI requests to services.
- `services_info_provider`  
/ *Condition*: required / *Type*: callable /  
Callable that returns the current services info dict.

#### 54.1.2 Method: stop

Stop the UI bridge server.

#### 54.1.3 Method: broadcast\_update

Push a services update to all connected UI clients.

**Arguments:**

- `services_info`  
/ *Condition*: required / *Type*: dict /  
Dict of all current service information.

# Chapter 55

## Appendix

### 55.1 Appendix

#### 55.1.1 License

MicroserviceBase is licensed under the Apache License, Version 2.0.

You may obtain a copy of the License at: <http://www.apache.org/licenses/LICENSE-2.0>

#### 55.1.2 References

- RabbitMQ: <https://www.rabbitmq.com>
- pika (RabbitMQ client for Python): <https://pika.readthedocs.io>
- FastAPI: <https://fastapi.tiangolo.com>
- Electron: <https://www.electronjs.org>
- Python: <https://www.python.org>

#### 55.1.3 Repository

Source code: <https://github.com/test-fullautomation/python-microservice-base>

Issue tracker: <https://github.com/test-fullautomation/python-microservice-base/issues>

#### 55.1.4 Contact

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# Chapter 56

## History

### 56.1 History

#### 56.1.1 Version 2.0.0 - 09.02.2026

Major restructure: hexagonal architecture, dual-host GUI, multi-broker support, local process hub, standalone installer.

##### Architecture

- **Hexagonal Architecture:** Reorganized codebase into domain, ports, and adapters layers
- **Factory Pattern:** All components created through `factory.py` for easy swapping
- **Port Interfaces:** `TransportPort`, `RegistryPort`, `UIBridgePort`
- **Adapter Implementations:** RabbitMQ transport, RabbitMQ registry, FastAPI bridge, Local Hub Manager, Service Executor

##### GUI v2.0

- **Dual Hosting:** Electron desktop app and browser mode via FastAPI bridge
- **Pure Web Frontend:** HTML/CSS/JS with Bootstrap 5 CDN (no Node.js dependencies)
- **Namespace:** `window.MicroserviceManager` (alias MM) replaces Node.js global and `require()` patterns
- **Service Plugins:** Dynamically loaded per-service UI components in `web/services/`
- **Dark Sidebar Theme:** Custom Bootstrap dark sidebar with accordion navigation
- **Login Modal:** Bootstrap modal replaces separate Electron popup window
- **Toast Notifications:** `MM.showToast()` replaces `notifier.notify()` and `dialog.showMessageBox()`

##### Multi-Broker Support

- **Connection Map:** `MM.connections` tracks per-broker state
- **Reverse Lookup:** `serviceToBroker` maps service names to broker endpoints
- **Progressive Disclosure:** Broker headers shown only with multiple connections
- **Session Persistence:** Connection state saved via `sessionStorage`

## Local Process Hub

- **Hub Manager:** Start, stop, and monitor local service processes
- **Service Executor:** Process execution with `CTRL_BREAK_EVENT` for graceful shutdown on Windows
- **Config Persistence:** `hub_processes.json` with `${python}` and `${config_dir}` placeholders preserved across save operations
- **Hub Dashboard:** Real-time process status display in the GUI
- **REST API:** Full hub management via `/api/local-hub/*` endpoints

## Service Management

- **Service Import:** Import microservice packages from folders or zip archives
- **Service Creator:** Interactive wizard for scaffolding new microservices
- **Registry Shutdown:** `__registry_shutdown__` sentinel notifies subscribers when a service unregisters

## Electron Packaging

- **Standalone Installer:** NSIS installer for Windows (`build.bat`), Linux packages (`build.sh`)
- **Read-only Resources:** Scripts in `resources/python/`, app in `app.asar`
- **Writable User Data:** Settings, config, services, logs in `%APPDATA%/devatservgui/`
- **First-Run Init:** Template files copied from resources on first launch
- **Bridge Lifecycle:** Detached bridge process managed via PID file

## Windows Process Fixes

- `SIGBREAK` handler converted to `KeyboardInterrupt` for graceful Python cleanup
- `Pika.start_consuming()` replaced with `process_data_events(time_limit=1)` loop for interruptible consumption
- `subprocess.PIPE` blocking prevented by using `FileHandler` instead of `StreamHandler` for logging in subprocess mode

## Documentation

- 12 Architecture Decision Records (ADR-001 through ADR-012)
- 12 PlantUML diagrams (overview, architecture, component, class, sequence, state)
- Updated README with architecture diagrams, quick start guide, and API reference
- Package documentation with Introduction, Description, and History

### 56.1.2 Version 0.1.0 - 02/2023

Initial version.

- Basic microservice framework with RabbitMQ communication
- Service registration and discovery
- Electron-based GUI for service monitoring

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